It is important to remember that there is no such thing as a perfectly secure system, and there is not even a single definition of “security.” For a healthcare provider, being able to discover whether your friends have accounts on a system would be considered a major security flaw and a privacy violation. However, for a social network the same capability is an essential feature. Security therefore depends on the context. There are many aspects that should be considered when designing a secure API, including the following:

- The *assets* that are to be protected, including data, resources, and physical devices.
- Which *security goals* are important, such as confidentiality of account names.
- The *mechanisms* that are available to achieve those goals.
- The *environment* in which the API is to operate, and the *threats* that exist in that environment.

1.4.1 Assets

For most APIs, the assets will consist of information, such as customer names and addresses, credit card information, and the contents of databases. If you store information about individuals, particularly if it may be sensitive such as sexual orientation or political affiliations, then this information should also be considered an asset to be protected.

There are also physical assets to consider such as the physical servers or devices that your API is running on. For servers running in a datacenter, there are relatively few